

G16 - Assignment 1

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Abstract—Time series forecasting aims to predict future values from historical data. In this study, we apply Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models to a laser measurement dataset. We tune key hyperparameters, including the learning rate and input sequence length, using Mean Squared Error (MSE) and Mean Absolute Error (MAE) as the evaluation metric.

Results show that both models effectively capture temporal patterns, with LSTM achieving slightly better performance. We further present the training process and 200-step-ahead forecasting results to demonstrate the effectiveness of the proposed approach.

I. INTRODUCTION

Time series forecasting is a fundamental problem in many real-world applications, where the goal is to predict future values based on historical observations. In this assignment, we focus on a real-life laser measurement dataset and aim to perform one-step-ahead prediction, as well as multi-step forecasting through recursive prediction. Such tasks are particularly challenging due to the presence of temporal dependencies, noise in the measurements, and the accumulation of prediction errors over time.

Traditional machine learning models often struggle to capture long-term dependencies in sequential data. To address this limitation, recurrent neural networks (RNNs), especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), have been widely used for time series modeling. LSTM is designed to capture long-range dependencies through memory cells, while GRU offers a more computationally efficient alternative with a simpler structure.

In this work, we explore both LSTM and GRU architectures for the given forecasting task. We investigate the impact of key hyperparameters, such as the number of past time steps used as input, and evaluate model performance using Mean Absolute Error (MAE) and Mean Squared Error (MSE). Furthermore, we assess the models' ability to generate long-term predictions by recursively forecasting future values and comparing them with the ground truth.

II. USED MODELS

A. Long Short-Term Memory (LSTM)

We selected Long Short-Term Memory (LSTM) networks as one of the models primarily because the laser measurements constitute sequential time series data. Unlike standard feedforward models that treat inputs independently, LSTMs

are explicitly designed to recognize temporal order and historical context. While standard RNNs struggle with long-range dependencies due to the vanishing gradient problem, LSTMs overcome this using specialized gating mechanisms. These gates allow the network to selectively retain relevant temporal trends, filter out measurement noise, and discard obsolete information over extended sequences, making them highly effective for capturing the complex, non-linear dynamics of physical time series signals.

B. Gated Recurrent Unit (GRU)

Gated Recurrent Unit (GRU) is a streamlined variant of Recurrent Neural Networks (RNNs) that, similar to LSTM, is engineered to address the vanishing gradient problem and capture temporal dependencies in sequential data. However, GRU offers a more computationally efficient architecture by consolidating the gating mechanisms. Instead of the three gates found in LSTM, the GRU utilizes two primary gates: the update gate and the reset gate. The update gate determines how much of the previous state should be retained, while the reset gate controls how much of the past information to forget when computing the new candidate state. This simplified structure often leads to faster training convergence while maintaining performance comparable to LSTM.

III. APPROACH

A. Long Short-Term Memory (LSTM)

To optimize the LSTM model, we performed a structured hyperparameter tuning procedure under a fixed data split, where 80% of the data was used for training and 20% for validation.

To control the complexity of the search space, we adopted a two-stage tuning strategy. In the first stage, we manually explored model capacity by testing combinations of hidden size 64, 128 and number of layers 1, 2, 3. Based on validation performance, the configuration with hidden size 64 and 2 layers was selected, as it achieved strong performance without introducing unnecessary model complexity.

In the second stage, we fixed these capacity-related parameters and conducted a grid search over the remaining hyperparameters. This design choice reduces the computational cost of exhaustive search while still allowing systematic exploration of key parameters.

The grid search was performed over the following ranges:

- Sequence length: {10, 15, 20, 25}

- Learning rate: $\{1e-4, 5e-4, 1e-3, 5e-3\}$

Each configuration was trained using a batch size of 32 for 100 epochs and evaluated on the validation set. Due to the relatively small size of the training data, we observed that batch size had a limited impact on performance, and therefore it was not included in the tuning process. After hyperparameter tuning, we also experimented with increasing the training partition from 80% to 95% to incorporate more recent temporal information. However, this modification did not lead to noticeable performance improvements for the LSTM model. Therefore, the original 80%/20% train-validation split was retained for the final experiments. The final configuration is summarized in Table I.

TABLE I
FINAL HYPERPARAMETER CONFIGURATION OF THE LSTM AND GRU MODELS

Hyperparameter	LSTM	GRU
Hidden Size	64	64
Number of Layers	2	1
Sequence Length	10	14
Learning Rate	0.001	0.001
Batch Size	32	64
Epochs	100	100

B. Gated Recurrent Unit (GRU)

In the initial phase, the raw X_{train} sequences are standardized to ensure numerical stability. To preserve the chronological structure of the time-series data, the data is split into a training set (80%) and a validation set (20%) without random shuffling. We perform a systematic Grid Search to optimize the window_size (ranging from 10 to 19) and the learning_rate (0.0001, 0.001, 0.01, 0.1). Recognizing that GRU models can be highly sensitive to stochastic initialization, each configuration is evaluated using three random seeds (42, 1, and 2). Furthermore, we implement a recursive forecasting approach to simulate long-term predictive performance across the entire validation set. The optimal hyperparameter combination is ultimately selected based on the lowest Average Recursive Mean Squared Error (MSE) across the three seeds.

With the optimal hyperparameters, the training partition is increased to 95% of the available data to incorporate more recent trend information, while the remaining 5% is reserved for a safety margin check. To mitigate the risk of numerical divergence or extreme biases inherent in a single model, an ensemble training strategy is executed, where three independent GRU models are trained with different random seeds. For the final 200-step recursive inference, the outputs from these three models are integrated using mean averaging (prediction smoothing).

IV. RESULTS

A. Long Short-Term Memory (LSTM)

To determine the optimal number of past time steps to input into the network (sequence length), we evaluated the

LSTM model's performance across various look-back windows ($k \in \{10, 15, 20, 25\}$). The performance for each configuration was quantified using the scaled Mean Squared Error (MSE) and Mean Absolute Error (MAE) on the validation set. The empirical results indicate that the model performs best with a sequence length of 10. Increasing the historical context beyond 10 steps did not provide further predictive benefits and instead resulted in a slight degradation of accuracy, likely due to the model struggling with added noise or irrelevant long-term historical dependencies. Consequently, a sequence length of 10 was selected for our final model evaluation.

Using the optimal hyperparameters identified during the grid search (Sequence Length = 10, Hidden Size = 64, Layers = 2, Learning Rate = 0.001), we trained the final LSTM model. The training process was monitored by evaluating the Mean Squared Error (MSE) loss on both the training and validation splits over 100 epochs. Figure 1 presents the learning curves for the optimized LSTM model. Both training and validation losses decrease rapidly during the initial epochs and quickly stabilize, demonstrating robust and efficient convergence. The close alignment between the training and validation loss curves throughout the training process confirms that the model generalizes well to unseen data and avoids significant overfitting.

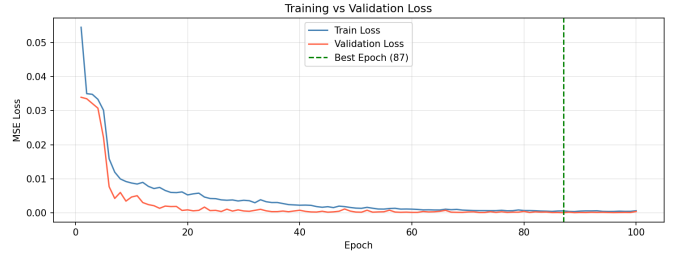


Fig. 1. Training and validation loss (MSE) over 100 epochs for the optimized LSTM model. The vertical dashed line indicates the epoch with the best validation performance.

Upon convergence, the optimized LSTM model was evaluated for its one-step-ahead prediction capabilities on the validation set. The performance metrics of the model are detailed in Table II. As shown, the model achieved high precision on the scaled data. When transformed back to the original scale of the real-world laser measurements, the model yielded a final MAE of 1.01, indicating that our predictions closely follow the underlying signal dynamics with minimal deviation.

TABLE II
ONE-STEP-AHEAD PREDICTION PERFORMANCE ON THE VALIDATION SPLIT OF THE LSTM AND GRU MODELS

Metric	LSTM	GRU
MAE (Original Scale)	1.01	1.19
MSE (Original Scale)	1.82	2.36

To evaluate the model's ability to generalize to completely unseen future data, we performed a recursive 200-step forecast and compared the predictions against the hold-out test dataset

(‘Xtest.mat’). Unlike one-step-ahead validation, recursive forecasting requires the model to feed its own predictions back as inputs for subsequent time steps, which typically leads to compounded errors over time. Figure 2 visually compares the model’s recursive forecast against the true ground truth values from the test set.

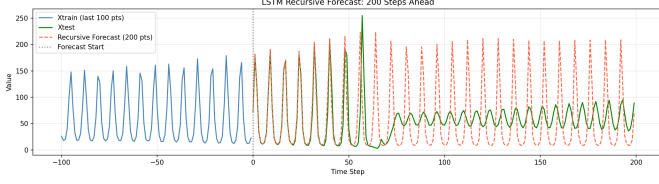


Fig. 2. Recursive 200-step forecast compared against the unseen ground truth test data (Xtest) on the LSTM model.

The quantitative performance metrics for this long-term recursive prediction are presented in Table IV. As expected, the error margins are significantly larger than those observed during one-step-ahead validation due to the inherent difficulty and error accumulation in long-term autoregressive forecasting.

TABLE III
RECURSIVE 200-STEP FORECAST PERFORMANCE ON THE UNSEEN TEST DATA (XTEST) OF LSTM

Metric	Value
MSE (Original Scale)	3877.12
MAE (Original Scale)	43.75

B. Gated Recurrent Unit (GRU)

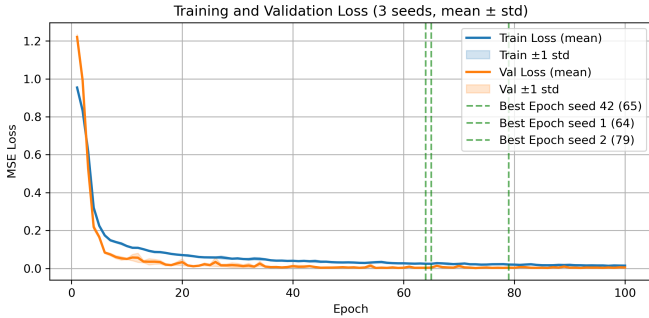


Fig. 3. Training and validation loss (MSE) over 100 epochs for the optimized GRU model. The vertical dashed line indicates the epoch with the best validation performance.

TABLE IV
RECURSIVE 200-STEP FORECAST PERFORMANCE ON THE UNSEEN TEST DATA (XTEST) OF GRU

Metric	Value
MSE (Original Scale)	3945.39
MAE (Original Scale)	50.92

Using the optimal hyper parameters from the grid search, we trained the final GRU model over 100 epochs. Figure 3

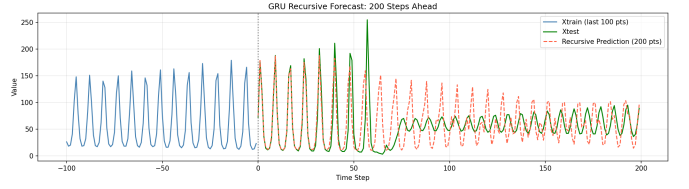


Fig. 4. Recursive 200-step forecast compared against the unseen ground truth test data (Xtest) on the GRU model.

shows the averaged training and validation loss curves across the three seeds, together with the standard deviation bands. Both losses decreased sharply during the first few epochs and gradually stabilized afterwards, indicating efficient convergence. The validation loss remained close to the training loss throughout most of the training process, which suggests that model did not have severe overfitting.

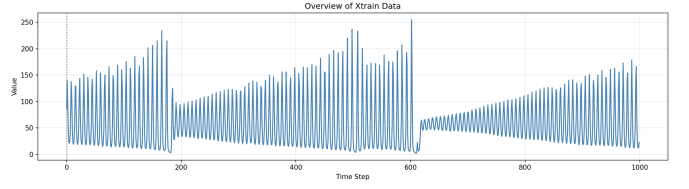


Fig. 5. The overview of train set.

As for the test result shown in figure 4, the model captures the short-term oscillatory pattern well in the early part of the prediction, indicating that the GRU model has learned the repeated short- period fluctuations in the signal. But after the sharp decrease around 60 time steps, the model continues to produce relatively strong periodic peaks.

From the overview of train set shown in figure 5, the signal contains both two periodic patterns. The short-term oscillations appears frequently, so the model can learn it well. In contrast, the long-term transitions, such as the sharp drops around time steps 190 and 610, occur only 2 times in the data. Therefore, the GRU has limited evidence to learn the full structure of these changes.

V. CONCLUSION

In this study, we investigated the performance of LSTM and GRU models for laser time-series forecasting. Both models achieved relatively good performance on learning the short-term oscillatory structure of the signal and getting low one-step-ahead validation errors. However, the recursive 200-step forecasts showed much larger errors, indicating that long-horizon prediction is more challenging due to error accumulation.

Overall, the LSTM achieved slightly better recursive forecasting MSE than the GRU on test set. Both of them struggled with rare long-term transitions and abrupt amplitude changes. This suggests that recurrent models’ ability to predict long-term pattern is limited when such patterns appear only a few times in the train set.